

Many-Analysts Simulation Study: Background and Scope of the Project

Background

Most psychological research relies on **convenience samples**, e.g. student samples, online opt-in panels, clinical samples, self-selected volunteers (Peterson & Merunka, 2014; Andrade, 2021; Wild et al., 2022). This raises a central question: when do results from non-representative samples generalize to the target population? Despite their ubiquity, convenience samples come with little guidance on the conditions under which their results generalize to the target population. This gap is problematic, as convenience sampling can undermine the replicability and generalizability of research findings.

Rationale and Meta-Methodological Motivation

Different choices in the analysis of research studies, also known as researcher degrees of freedom (Simmons et al., 2011) (RDF), and the possibility of (unintentionally) exploiting them to obtain desired results, have been identified as one factor contributing to the replication crisis in science. Many-analysts studies, in which multiple independent research teams or analysts address the same research question on the same data (Silberzahn et al., 2018; Aczel et al., 2021). Rather than treating one analysis as definitive, they quantify how conclusions vary across reasonable analytical choices.

The same idea can be transferred to simulation-based methodological research. In simulation studies, analysts can share the same research question and target estimands while varying selected RDF in the simulation design. For example, the lead team may fix the research question, the competing methods, and the analysis targets, while analysts independently specify the DGP and relevant implementation choices. In this way, a many-analysts simulation study provides a pluralistic perspective on broad methodological questions and helps assess the sensitivity of conclusions to alternative, yet plausible, design choices (Aczel et al., 2021; Sarafoglou et al., 2024, 2025).

In our case study, we compare different effect size measures with respect to their robustness to non-representative sampling mechanisms. Quantifying variability in this context provides a first assessment of variability in simulation-based methodological research in settings that reflect typical practice, where multiple analysts independently make design and analysis choices within a shared framework. Our study provides practical insights into the operational feasibility of many-analysts simulation studies as a tool.

Case study: Identifying conditions under which the results in convenience samples generalize to the target population

The aim of this project is to assess the degree to which non-representative sampling, operationalized as either (1) restriction in the predictor X or (2) restriction in the outcome Y , affects the following common effect size measures: the unstandardized and standardized regression coefficients (β_1 , β_1^{std}), the Pearson correlation coefficient (r), Cohen’s d (d), and the coefficient of determination (R^2). To this end, we evaluate two hypotheses, each corresponding to one type of restriction and a third hypothesis corresponding to power. Assuming a correctly specified multiple linear model, we formulate the following three hypotheses:

Hypothesis 1: When the sampling mechanism restricts the range or distribution of the predictor X , the unstandardized regression coefficient β_1 and Cohen’s d remain unbiased, whereas β_1^{std} , R^2 and r are in general biased.

Example: If one relies on a sample of university students to estimate the association between intelligence and creativity, the range of intelligence as predictor may be changed because on average university students tend to score higher on intelligence tests than the general population.

Hypothesis 2:

When the sampling mechanism restricts the range or distribution of the outcome Y , all five effect size measures are in general biased.

Example: If one relies on a clinical sample to assess the extent to which trauma and stressful life events influence mental health, then the outcome (mental health) is changed in its range, because patients tend to have poorer mental health than the general population.

Hypothesis 3: Restricting the range or distribution of the predictor X or the outcome Y will lead to a loss of power.

For either scenario – range restrictions in X or Y – we expect the bias in the estimated effect sizes to depend on the severity of the restriction, with more extreme restriction yielding more biased results. Crucially, range restriction does not merely change the observed data; it causes the effect-size estimators themselves to become biased, meaning that the expected estimate no longer equals the true population effect size. The magnitude of this estimator bias depends on both the degree of range restriction and the underlying strength of association between X and Y . In particular, we generally would expect range restriction in X or Y to lead to no bias in any effect size measures in the case of X and Y being uncorrelated.

Identifying when and how non-representative sampling mechanisms can generalize to a target population requires both nuance and methodological pluralism. For this reason, our many-analysts simulation study includes a confirmatory and an exploratory component. In the confirmatory phase, analysts work within a predefined and restricted set of RDF, specifying only one data-generating process (DGP), which is executed via a standardized wrapper function. In the exploratory phase, analysts may choose from a broader range of RDF, enabling a more open investigation of the hypotheses and revealing how different analytical choices influence the conclusions.

Analyst Involvement and Surveys

Participants will receive a detailed instruction document and a template R script. They are asked to:

- implement a data-generating process (DGP) within predefined constraints,
- generate simulated datasets under different restriction scenarios,
- run a provided wrapper function to compute predefined effect-size measures,
- and submit the resulting code and output to the research team.

The task requires experience in simulation-based statistical programming in R. No sensitive data are used; all data are simulated.

Possible risks and benefits

There are no direct risks associated with the participation in the study. Analysts who make a substantial scientific contribution to the study (e.g., through the development of the DGP, engagement in manuscript development, and approval of the final version) may be invited to become co-authors in accordance with established authorship guidelines (e.g., DFG Code of Conduct). Co-authorship is not granted automatically through participation and is not intended as compensation, but reflects substantive intellectual contribution and responsibility.

Expected time commitment

The expected time investment for participation in the simulation component is approximately 12–20 hours. This includes:

- familiarisation with the study instructions,
- implementation of the DGP within predefined constraints,
- execution of the wrapper function,
- optional exploratory simulation analyses, and

- minor revisions if required for technical consistency

The time required may vary depending on prior experience with simulation studies.

If an analyst wishes to be considered for co-authorship, additional engagement in manuscript preparation is required. This may involve approximately 10–15 additional hours, including:

- reviewing and revising draft versions of the manuscript,
- contributing to interpretation and discussion,
- responding to reviewer comments where appropriate, and
- approving the final manuscript version.

Participation in manuscript preparation is voluntary and not required for completion of the simulation study. Participation can be completed flexibly and independently within the specified timeframe.

Next steps

Researchers interested in joining the many-analysts simulation study are invited to contact us at multi.analyst@stat.uni-muenchen.de. Upon contacting us, you will receive all relevant materials, including scripts and your unique analyst identification number. Please direct any further questions to the same email address.

References

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